



Society of Petroleum Engineers

SPE-227414-MS

Sequential Plugging Fluid Modeling for Lost-Circulation in Fractured Formations

R. Albattat, Sinopec Tech Middle East LLC, Dhahran, Saudi Arabia; X. He, Saudi Aramco PE&D EXPEC ARC, Dhahran, Saudi Arabia; Y. Li and H. Hoteit, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227414-MS](https://doi.org/10.2118/227414-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Summary

This paper investigates the application of type-curves for mud loss incidents within a natural fracture. The established solution mimics and predicts drilling mud loss in subsurface formations with conductive natural fractures. Mud loss during drilling is a complex and costly issue that disrupts operations and increases non-productive time. The goal of this study is to develop a transient solution tool that leverages mud type-curves during operational parametric variations to model multi-stage sequential injection of lost circulation material (LCM). This curve will provide key parameters such as transient shift mud loss volume, transient shift mud loss duration.

The study integrates a semi-analytical approach to model mud loss using non-Newtonian fluid model, i.e. the Herschel-Bulkley fluid model, which accounts for shear-thinning behavior. A Cauchy equation of motion has been used to establish a tool that takes input parameters, such as yield stress, consistency factor, flow behavior index, wellbore size, bottom-hole pressure, and hydraulic fracture aperture, to generate a shift-trend solution. In current literature, the approach typically utilizes an analytical solution with constant parameters such as the fluid and formation properties, yet these properties may change over time. To address this challenge, we introduce a novel concept of shift-trend solution caused by variational parameters during drilling operations to measure and predict mud loss dynamics. The model was established and validated using field data to ensure accuracy and adaptability.

The developed shift-trend model demonstrated high accuracy in predicting the dynamic trend with dynamic mud loss volume, maximum loss duration, and equivalent hydraulic fracture aperture under a different range of conditions. It effectively captured the complex, nonlinear relationships governing mud loss behavior in naturally fractured formations. The shift-trend solution successfully integrated Cauchy equation of motion, yielding predictions that are closely aligned with field observations. This streamlined approach reduces computational demands while maintaining reliability, offering practical solutions for real-time decision-making in lost circulation scenarios.

This study introduces a novel shift-trend solution framework for modeling and mitigating mud loss in naturally fractured formations. By combining non-Newtonian fluid model with Cauchy equation of motion,

the proposed tool enhances the predictive capabilities of traditional methods and provides actionable insights for managing lost circulation via additives to stop/mitigate lost circulation severity, such as lost-circulation materials. The approach is adaptable to diverse scenarios, making it an accurate and efficient solution for addressing one of the oil and gas industry's most persistent challenges.

Introduction

As the oil and gas industry is striving to improve efficiency and reduce operational cost, lost-circulation (LC) during drilling remains a major issue, disrupting drilling operations across many regions worldwide, particularly in carbonate formation. It is defined as the uncontrolled escape of wellbore fluid into the surrounding formation without return, occurring during drilling, cementing, or workover. This phenomenon significantly increases non-productive time, operational costs, and safety risks to personnel [1]. LC can occur at any stage of well construction, including drilling, cementing, or workover operations. At some critical cases, loss incidents may appear unexpectedly and cannot be anticipated from offset wells [2]. LC is also common and particularly hazardous in Antarctic drilling, where subsurface ice layers are penetrated [3].

Many authors have investigated the problem from multiple perspectives to either eliminate or mitigate the loss. Lost-circulation materials (LCM) are widely developed and implemented in the field to stop or reduce the loss incidents, including cement slurry [4], biomass materials [5], network-like materials [6], Antarctic low-temperature plugging agent [3], ultra-deep LCM control [7], reactive-worm-like materials [8], and consolidation plugging agent's [9]. Others have studied the loss from modeling aspects to gain insights into the nature of the loss and acquire some formation properties, such as mean aperture size of natural fracture [10], [11]. Estimation of fracture aperture is a pivotal criterion to design and deploy LCM as each type is intended to be effective within specific range of fracture width [12]. As a result, type-curves have been introduced as key tools to predict average hydraulic aperture and cumulative fluid loss volume over time, either through uncertainty analysis [13], or by using machine learning methods [14]. Geomechanical solutions were provided to model compression of the LCM plug upon closure of an inclined fracture faces and to model spatial distribution of mud losses [15], [16]. For real-time, LC alarming system was addressed to provide a system that proactively warns the drillers [17], while others have proposed a need for more accurate forecasting tools [18]. Combining seismic data using machine learning was attempted consequently to enhance prediction accuracy of potential mud loss incidents by using machine learning [19]. Proposing interpretable lost circulation analysis framework has been addressed based on machine learning for labeling, types and categorizing goals of LC [20].

The presence of natural fractures, which are inherently conductive to fluid flow, makes formations prone to lost circulation (LC). This issue is especially prevalent in carbonate formations, notably in the Middle East region [21]. Further, the issue exacerbates in geothermal field development, reaching up to 30% of the drilling costs [22]. Time plays a critical role in the problem as fluid losses without expectancy plus adding another layer of complexity due to complicated subsurface formations along with varying drilling parameters [23]. Thus, practical solutions adhering to quick response in real-time analysis are crucial to actively mitigating the loss. Corrective measures are also required simultaneously with prediction solutions to provide recommendations to reduce the loss under acceptable level. Dynamic changes to the circulating fluid can reduce or induce stopping of the loss due to thicker fluid properties, since non-Newtonian fluid is superior in those cases. In this paper, the proposed solution mimics a non-Newtonian fluid invading into conductive natural fracture and modeling LC. Then, a plugging fluid containing LCM is going to be introduced at certain timepoint entering the fracture and gradually affecting the losing fluid to slow down until a complete halt is reached. Different properties of the plugging fluid or invading fluid cause the loss rate to vary accordingly. This work, for the first time, has been introduced in literature as a form of semi-analytical solution, where two non-Newtonian fluids invade into the fracture, plugging fluid influencing the invading fluid.

The paper is structured into main sections. The introduction of the main problem is depicted and expressed as regards to the necessity of a practical solution for LC given the time nature of the incident. Then, physical model development is shown to render the final solution to be used in real-time analysis. Afterward, a demonstration of the solution scenarios is illustrated to show the sensitivity of lost fluid along with varying the properties of the plugging fluid. Finally, some published field cases are utilized for feasibility of the solution and to show simplicity and effectiveness for the proposed solution. Finally, remarkable points are deduced from this study.

Modeling Aspect

Problem Formulation

The objective is to develop a model that represents corrective measures applied after a lost circulation (LC) event has occurred. The problem is first approached by modeling the initial fluid encroachment into a natural fracture, characterizing it as an LC phenomenon. Subsequently, a specially formulated plugging fluid is introduced as a corrective measure to mitigate fluid loss and reduce the total duration of the LC event. Fig. 1 depicts lost mud being displaced by a "thicker" plugging fluid, whose properties are reshaped specifically to combat LC. Due to the presence of two-phase flow, both fluid fronts are being tracked dynamically with time, as the invasion advances further deep into the fracture. The inlet demonstrates the wellbore, and the two plates represent the upper and lower of a fracture system, see Fig. 2. Each fluid exhibits distinct shear stress and velocity profiles along the radial direction. As both fluids advance, the height of the yield stress region increases, marking the criteria for LC mitigation and eventual stoppage.

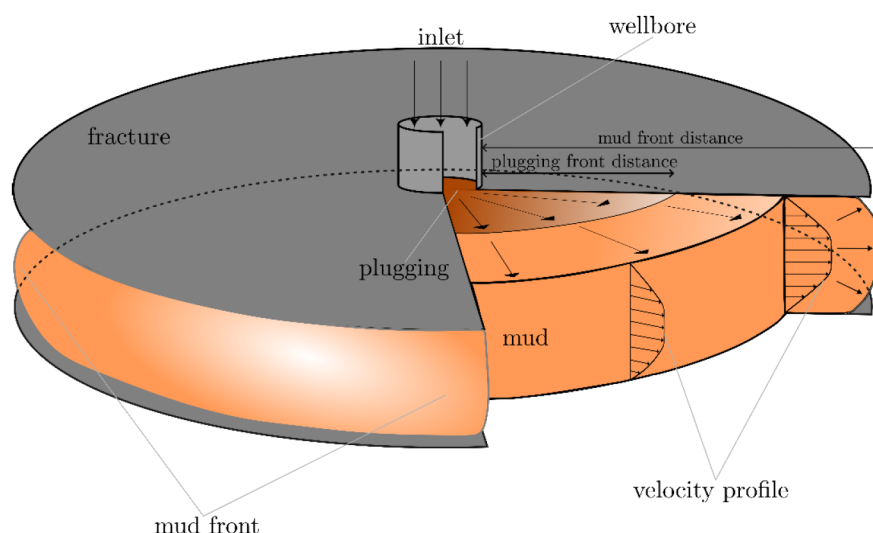


Figure 1—Schematic illustration of the lost circulation scenario showing two-phase flow within a natural fracture. A high-viscosity plugging fluid is injected from the wellbore and progressively displaces the previously lost drilling mud. The figure highlights the dynamic fronts of both fluids, along with their respective velocity profiles, as they advance radially through the fracture.

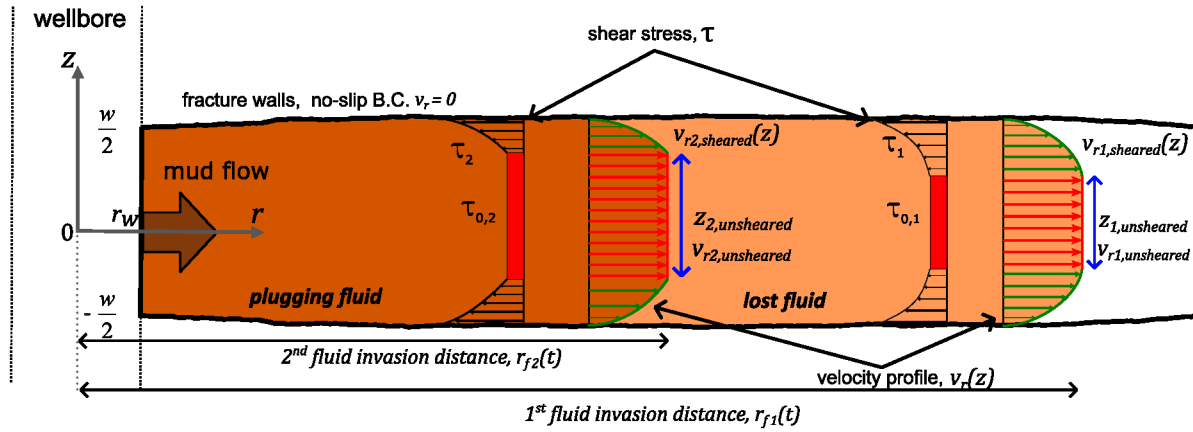


Figure 2—Cross-sectional view of the fracture showing the velocity and shear stress profiles for two non-Newtonian fluids during lost circulation control. The diagram illustrates the displacement of the initial mud by a plugging fluid, with each fluid characterized by distinct yield stress ($\tau_{0,1}$, $\tau_{0,2}$) and velocity profiles. The transition from unsheared to sheared zones is highlighted, demonstrating the rheological behavior of Herschel-Bulkley fluids along the fracture height.

Rheology and Cauchy of Motion

There are three main modeling stages to achieve the complete set of fluid flow dynamics. The first aspect is fluid model itself, i.e., rheological model. The fluid is required to be addressed in terms of shear stress under different shear rates. The optimum fluid model describing the drilling fluid is Herschel-Bulkley fluid model. It is described by three fluid properties; flow behavior index n , consistency multiplier factor, m , and fluid yield stress, τ_0 , as;

$$\tau(z, r) = \tau_0 + m \left(\frac{dv_r}{dz} \right)^n \quad (1)$$

Eq. (1) describes the shear stress, $\tau(z, r)$, varying with fluid height perpendicular to the flow inside the fracture, z , and radial distance, r , along the flow direction exerted on the fluid initiated from the fracture upper and lower faces. The shear rate is denoted as $\frac{dv_r}{dz}$, where the radial velocity, v_r , perpendicular to the flow direction is varying along the elevation height. The link between shear stress and shear rate is the central point of rheology. The benefits of Herschel-Bulkley model, represented in Eq. (1), can be reduced to three fluid models: Bingham-Plastic model, Power-Law model and Newtonian model. In case of Newtonian fluid model, the relationship between shear rate and stress has a straight-line behavior, governing by one parameter, i.e., viscosity. Similar behavior is established with Bingham-Plastic model as a straight-line, differently with shifted y-axis with a fixed value, i.e., fluid yield stress. Thus, this fluid type will remain unsheared until a minimum value (fluid yield stress) is exceeded to the fluid. In addition, Power-law fluids consist of "no fluid yield stress parameter $\tau_{0=0}$ ". It manifests non-linearity behavior controlled by flow behavioral index, where $n > 1$ demonstrates shear-thickening and $n < 1$ expresses a shear-thinning fluid. Indeed, $n = 1$ reduces to Newtonian model.

The second aspect is the movement of fluids in space and time. The main governing equation properly describing the fluid dynamics of non-Newtonian fluid is Cauchy equation of motion [24],

$$\rho \frac{\partial v}{\partial t} + \rho (v \cdot \nabla) v = \nabla \cdot (-pI + \tau) \quad (2)$$

Under the assumption of Stokes flow, the transient effect, $\rho \frac{\partial v}{\partial t}$, where velocity, v , changes with respect to time, t , is small since laminar is expected. The fluid shear stress term, τ , and the pressure force with identity matrix, pI are much higher than the convective term, which is eliminated from Eq. (2). Therefore, the governing terms are,

$$\nabla \cdot (-p\mathbf{I} + \boldsymbol{\tau}) = 0 \quad (3)$$

Applying polar coordinate system to avail from axisymmetric property of the flow, Eq. (3) gives,

$$\tau(z, r) = z \frac{\partial p}{\partial r} \quad (4)$$

Since the two main equations, fluid model and flow model, are generate, combining Eq. (4) and Eq. (3) renders,

$$\tau_0 + m \left(\frac{dv_r}{dz} \right)^n = z \frac{\partial p}{\partial r} \quad (5)$$

The aim is to solve radial velocity, v_r , as a function of elevation height and radial distance, as,

$$v_{r,sheared}(z) = \frac{n}{n+1} \left(z_{unsheared} - \frac{w}{2} \right) \left(\frac{\frac{\partial p}{\partial r} \left(\frac{w}{2} - z_{unsheared} \right)}{m} \right)^{\frac{1}{n}} + \frac{n}{n+1} (z - z_{unsheared}) \left(\frac{\frac{\partial p}{\partial r} (z - z_{unsheared})}{m} \right)^{\frac{1}{n}} \quad (6)$$

$$v_{r,unsheared} = \frac{n}{n+1} \left(\frac{\tau_0}{\frac{\partial p}{\partial r}} - \frac{w}{2} \right) \left(\frac{\left(\frac{w}{2} \frac{\partial p}{\partial r} - \tau_0 \right)}{m} \right)^{\frac{1}{n}}$$

Eq. (6) was derived by setting reference coordinates system at the midpoint of fracture aperture, i.e. the boundary is at $\pm w/2$. The Boundary Condition (BC) is assumed to be no-slip BC at the fracture wall, w , at where the velocity is zero, $v_r(\pm w/2) = 0$. The interface, between unsheared velocity region, $v_{r,unsheared}$, which has a constant velocity profile and sheared velocity region, $v_{r,sheared}(z)$, which has a variable velocity profile along the elevation height, is denoted as $z_{unsheared}$. As a result, sheared velocity is bounded by $Z_{unsheared} < Z < \frac{w}{2}$ and unsheared velocity is limited by $0 \leq z \leq z_{unsheared}$.

Mass Conservation Equations

The application of mass conservation is beneficial to amalgamate the two velocity profiles, i.e. sheared velocity profile and unsheared velocity profile. Recalling the mass conservation,

$$m_{in} - m_{out} = m_{acc} \quad (7)$$

Since the mass is entering the system, m_{in} , no mass is leaving the system, $m_{out=0}$ and the mass in the system, m_{acc} , is accumulated, then, Eq. (7) simplifies to,

$$m_{in} = m_{acc} \quad (8)$$

From density definition, ρ , fluid volume, V , can be addressed,

$$(\rho V)_{in} = (\rho V)_{acc} \quad (9)$$

Since the fluid is incompressible, density, ρ , is constant during the flow in space and time, Eq. (9) gives,

$$(V)_{in} = (V)_{acc} \quad (10)$$

Looking into volume accumulation in the fracture system whose entry is at $r=r_w$ and the fluid front is at $r=r_f$, the accumulated fluid geometric volume has a circular shape following an area of a circle extended in elevation height by the size of fracture aperture, w . Thus, Eq (10) becomes,

$$(V)_{in} = w\pi(r_f^2 - r_w^2) \quad (11)$$

As controlled volume is entering the system dynamically with respect to time, Eq. (11) is differentiated accordingly,

$$\frac{d(V)_{in}}{dt} = \frac{d}{dt} \left(w\pi(r_f^2 - r_w^2) \right) \quad (12)$$

The total volumetric flow rate, Q_{tot} , enters the fracture at the entry fracture face from the wellbore, and continues to invade far deep into the fracture. The only term changes dynamically as the fluid flows is fluid front. Thus, Eq. (12) is rearranged and solved for the derivative,

$$Q_{tot} = 2w\pi r_f \frac{dr_f}{dt} \quad (13)$$

Fluid Flow Equation

The total volumetric flow rate comprises sheared and unsheared velocity profiles and can be written as,

$$Q_{tot} = Q_{unsheared} + Q_{sheared} \quad (14)$$

From geometry of the fluid, total volumetric flow rate equals to the sum of both curved surface areas of rectangle occupied by the two velocity profiles multiplied by their corresponding velocity,

$$Q_{tot} = 2\pi r (2z_{unsheared}) v_{r,unsheared} + 2\pi r \left(2 \int_{z_{unsheared}}^{w/2} v_{r,sheared}(z) dz \right) \quad (15)$$

The integral term emerges because of the variability of sheared velocity profile, unlike unsheared one whose velocity remains unchanged with respect to elevation height. Substituting Eq. (6) into Eq. (15),

$$Q_{tot} = \frac{4\pi r}{m^{1/n}} \left(\frac{dp}{dr} \right)^{1/n} \left(\frac{w}{2} \right)^{\frac{1}{n}+2} \left(\frac{n}{2n+1} \right) \left(1 - \frac{\tau_0}{\frac{w}{2} \frac{dp}{dr}} \right)^{1/n} \left(1 - \left(\frac{1}{n+1} \right) \frac{\tau_0}{\frac{w}{2} \frac{dp}{dr}} - \left(\frac{n}{n+1} \right) \left(\frac{\tau_0}{\frac{w}{2} \frac{dp}{dr}} \right)^2 \right)^2 \quad (16)$$

Raising both sides to a flow behavior index power, rewriting Eq. (16) gives,

$$Q_{tot}^n = \frac{(4\pi r)^n}{m} \left(\frac{dp}{dr} \right) \left(\frac{w}{2} \right)^{2n+1} \left(\frac{n}{2n+1} \right)^n \left(1 - \frac{\tau_0}{\frac{w}{2} \frac{dp}{dr}} \right)^n \left(1 - \left(\frac{1}{n+1} \right) \frac{\tau_0}{\frac{w}{2} \frac{dp}{dr}} - \left(\frac{n}{n+1} \right) \left(\frac{\tau_0}{\frac{w}{2} \frac{dp}{dr}} \right)^2 \right)^{2n} \quad (17)$$

Taylor series is applied on the last term of the right side. The above equation is simplified to,

$$Q_{tot}^n = \frac{(4\pi r)^n}{m} \left(\frac{dp}{dr} \right) \left(\frac{w}{2} \right)^{2n+1} \left(\frac{n}{2n+1} \right)^n \left(1 - \left(\frac{2n+1}{n+1} \right) \frac{\tau_0}{\frac{w}{2} \frac{dp}{dr}} \right) \quad (18)$$

Expressing explicitly the pressure term in the above equation,

$$-\frac{dp}{dr} = \frac{m Q_{tot}^n}{\left[4\pi r \left(\frac{n}{2n+1} \right) \left(\frac{w}{2} \right)^{2n} \right] \left(\frac{w}{2} \right)} + \left(\frac{2n+1}{n+1} \right) \left(\frac{2\tau_0}{w} \right) \quad (19)$$

Superposition Principle

The modeling of two-phase non-Newtonian fluid flow is approached using the principle of superposition in time. The corrective measures for lost circulation (LC) begin with the introduction of a plugging fluid (referred to as the second fluid), which invades the fracture and dynamically displaces the previously lost drilling mud (the first fluid). As a result, two distinct fluid fronts are present within the fracture system, each advancing over time.

Since there are two-phase fluids flowing in the fracture system, the left-side integration of Eq. (19) ranges from the wellbore pressure, P_w , to the interface pressure, P_{f2} , (between 2nd fluid & 1st fluid), and then from the interface pressure to the 1st fluid front pressure, P_{f1} . Reflecting similar operation on the right-side of Eq. (19), the integration limits from wellbore radius to the 2nd fluid front, r_{f2} , and then from 2nd fluid front to 1st

fluid front, r_{f1} . In summary, the integration contains the span of first fluid and second fluid, whose fronts are dynamically changing with time. Mathematically, this dynamic interaction is expressed as:

$$\int_{P_w}^{P_{f2}} dp + \int_{P_{f2}}^{P_{f1}} dp = \int_{r_w}^{r_{f2}(t)} P_2(r) dr + \int_{r_{f2}(t)}^{r_{f1}(t)} P_1(r) dr \quad (20)$$

Applying this concept to Eq. (19),

$$\begin{aligned} (P_w - P_{f2}) + (P_{f2} - P_{f1}) &= \frac{m_2 Q_{tot}^{n_2} \left(r_{f2}^{1-n_2} - r_w^{1-n_2} \right)}{\left[4\pi \left(\frac{n_2}{2n_2+1} \right) \left(\frac{w}{2} \right)^{2n_2} \right]^{n_2} (1-n_2) \left(\frac{w}{2} \right)} + \left(\frac{2n_2+1}{n_2+1} \right) \left(\frac{2\tau_{02}}{w} \right) (r_{f2} - r_w) \\ &+ \frac{m_1 Q_{tot}^{n_1} \left(r_{f1}^{1-n_1} - r_{f2}^{1-n_1} \right)}{\left[4\pi \left(\frac{n_1}{2n_1+1} \right) \left(\frac{w}{2} \right)^{2n_1} \right]^{n_1} (1-n_1) \left(\frac{w}{2} \right)} + \left(\frac{2n_1+1}{n_1+1} \right) \left(\frac{2\tau_{01}}{w} \right) (r_{f1} - r_{f2}) \end{aligned} \quad (21)$$

Recalling Eq. (13), the total volumetric flow rate manifests displacement of control volume over time within the system. Due to incompressibility, total volumetric flow rate can be calculated as a rate of change of a control volume. Therefore,

$$Q_{tot} = \frac{dV}{dt} = 2\pi w r_{f1} \frac{dr_{f1}}{dt} = 2\pi w r_{f2} \frac{dr_{f2}}{dt} \quad (22)$$

Substituting Eq. (22) into Eq. (21),

$$\begin{aligned} P_w - P_{f1} &= \frac{m_2 \left(2\pi w r_{f1} \frac{dr_{f1}}{dt} \right)^{n_2} \left(r_{f2}^{1-n_2} - r_w^{1-n_2} \right)}{\left[4\pi \left(\frac{n_2}{2n_2+1} \right) \left(\frac{w}{2} \right)^{2n_2} \right]^{n_2} (1-n_2) \left(\frac{w}{2} \right)} + \left(\frac{2m_2+1}{m_2+1} \right) \left(\frac{2\tau_{02}}{w} \right) (r_{f2} - r_w) \\ &+ \frac{m_1 \left(2\pi w r_{f1} \frac{dr_{f1}}{dt} \right)^{n_1} \left(r_{f1}^{1-n_1} - r_{f2}^{1-n_1} \right)}{\left[4\pi \left(\frac{n_1}{2n_1+1} \right) \left(\frac{w}{2} \right)^{2n_1} \right]^{n_1} (1-n_1) \left(\frac{w}{2} \right)} + \left(\frac{2m_1+1}{m_1+1} \right) \left(\frac{2\tau_{01}}{w} \right) (r_{f1} - r_{f2}) \end{aligned} \quad (23)$$

The above equation is semi-analytical, highly non-linear implicit Ordinary Differential Equation (ODE) that imitates fluid flow of two-phase non-Newtonian fluid following Herschel-Bulkley rheology along an equivalent average hydraulic aperture where one fluid displaces another fluid acting as piston-like. The corrective measure is taken after the incident has occurred and an interval of time is passed, i.e. bounded by $t \rightarrow (0, t_p)$. At this duration, one-phase flow is therefore expected to flow. Applying $r_{f2} = r_w$ to Eq. (23),

$$P_w - P_{f1} = \frac{m_1 \left(2\pi w r_{f1} \frac{dr_{f1}}{dt} \right)^{n_1} \left(r_{f1}^{1-n_1} - r_w^{1-n_1} \right)}{\left[4\pi \left(\frac{n_1}{2n_1+1} \right) \left(\frac{w}{2} \right)^{2n_1} \right]^{n_1} (1-n_1) \left(\frac{w}{2} \right)} + \left(\frac{2m_1+1}{m_1+1} \right) \left(\frac{2\tau_{01}}{w} \right) (r_{f1} - r_w) \quad (24)$$

After corrective measures are deployed, a second phase of fluid is introduced to the system. Eq. (23) is applied within the time domain $t \rightarrow (t_p, t_{stop})$. For proper reading of Eq. (23), coefficients for each fluid are constructed as follows,

$$\begin{aligned}
\Delta P &= P_w - P_{f1} \\
B_1 &= \frac{m_1(2\pi w)^{n_1}}{\left[4\pi\left(\frac{n_1}{2n_1+1}\right)\left(\frac{w}{2}\right)^2\right]^{n_1} (1-n_1)\left(\frac{w}{2}\right)} \\
B_2 &= \frac{m_2(2\pi w)^{n_2}}{\left[4\pi\left(\frac{n_2}{2n_2+1}\right)\left(\frac{w}{2}\right)^2\right]^{n_2} (1-n_2)\left(\frac{w}{2}\right)} \\
C_1 &= \left(\frac{2n_1+1}{n_1+1}\right)\left(\frac{2\tau_{01}}{w}\right) \\
C_2 &= \left(\frac{2n_2+1}{n_2+1}\right)\left(\frac{2\tau_{02}}{w}\right)
\end{aligned} \tag{25}$$

Replacing coefficients of Eq. (25) with Eq. (23),

$$\Delta P = B_2 \left(r_{f1} \frac{dr_{f1}}{dt} \right)^{m_2} \left(r_{f2}^{1-m_2} - r_w^{1-m_2} \right) + C_2 (r_{f2} - r_w) + B_1 \left(r_{f1} \frac{dr_{f1}}{dt} \right)^{m_1} \left(r_{f1}^{1-m_1} - r_{f2}^{1-m_1} \right) + C_1 (r_{f1} - r_{f2}) \tag{26}$$

Solution Approach

Auxiliary conditions must be first addressed as Initial Value Problem (IVP) for the ODE. As a result, the steps to solve the problem are divided into two stages: one phase flow and two-phase flow domains. At single phase, LC is occurring dynamically and only one fluid is considered and solved. Thus, the time domain within this scheme is $t \rightarrow (0, t_p)$ where zero notates for onset of LC and t_p denotes the timepoint of a second fluid commencement. Afterwards at two-phase, corrective fluid has been introduced in the system at time t_p and consequently the two fluids continue to flow until stationary conditions reached, ranging over time domain $t \rightarrow (t_p, t_{end})$. The solution steps are as follow;

1. Apply initial conditions for 1st fluid, at initial time, $t=0$, the fluid front is $r=r_w=r_{f1}$
2. Identify the numerical value of injection time for second fluid, t_p
3. Solve 1st fluid front distance, r_{f1} , limited within a time range $t \rightarrow (0, t_p)$ using Eq. (24) and initial condition mentioned in step#1, while knowing fluid properties, average fracture aperture, pressure drop and wellbore size. This can be done numerically via any scheme, for example Runge-Kutta-Fehlberg method (RK45). The output solution has two arrays of 1st fluid front radii distance and its corresponding time.
4. Extract the solution point of $r_{f1}(t_p)$ and use it as an initial condition for the second stage.
5. Apply initial condition for 2nd fluid, at initial time, $t=t_p$, the fluid front is $r=r_w=r_{f2}$
6. Identify a stopping time, t_{end} , where can be captured by imposing a rule, i.e. t_{end} at $\frac{dr_{f1}}{dt} \approx 0$
7. At a tiny new timestep after t_p , solve 1st fluid front distance, r_{f1} , based on the previous timestep of the fluid front as mentioned in step#4, using Eq. (23) and also by applying the initial condition of the 2nd fluid mentioned in step#5 for the previous timestep. The 2nd fluid properties should be known. At the same new timestep, the fluid front, r_{f1} , can be found by using root-finding algorithm such as Newton-Raphson method, since Eq. (23) becomes an implicit non-linear ODE.
8. At the same timestep, find 2nd fluid front, r_{f2} , by using Eq. (22) and apply the conditions mentioned in step#5, step#4 and the solution found in step#7.
9. Repeat again for each timestep for the next successive timesteps as mentioned in step#7 and step#8 since the new timesteps will become now the old timesteps.
10. For practical purposes, calculation of cumulative fluid loss volume can be done using Eq. (11) since the cumulative volume is more practical than fluid front distance.

Solution Demonstration

Visualizing the solution is essential for assessing its feasibility and practical applicability. Once the solution is obtained, the results can be portrayed as 2D plot or dynamic animation. Fig. 3 shows an example scenario where LC happens into natural fracture body, including corrective measures. X-axis shows an argument variable which is the elapsed time in sec unit as the lost fluid has entered the fracture until stoppage, while y-axis exhibits the output that is cumulative fluid loss volume in $[m^3]$ unit. The blue graph is the track of cumulative loss volume for both fluids, whereas the orange one is track of losing volume of 2nd fluid, which is the corrective fluid. The dashed red line is the timepoint where onset of the corrective fluid is introduced and, in this scenario, it is equal to $t_{p=20} [mins]$. The figure is plotted on log-log scale. The annotations in green color mark the calculated final cumulative loss volume at stopping conditions and the halting time is $t_{end=15.5} [hrs]$. The total cumulative loss volume for both fluids at halt condition is 18.1 $[m^3]$, consisting of the 1st fluid cumulative lost volume which is 3.8 $[m^3]$ and 2nd fluid cumulative lost volume which is 14.3 $[m^3]$.

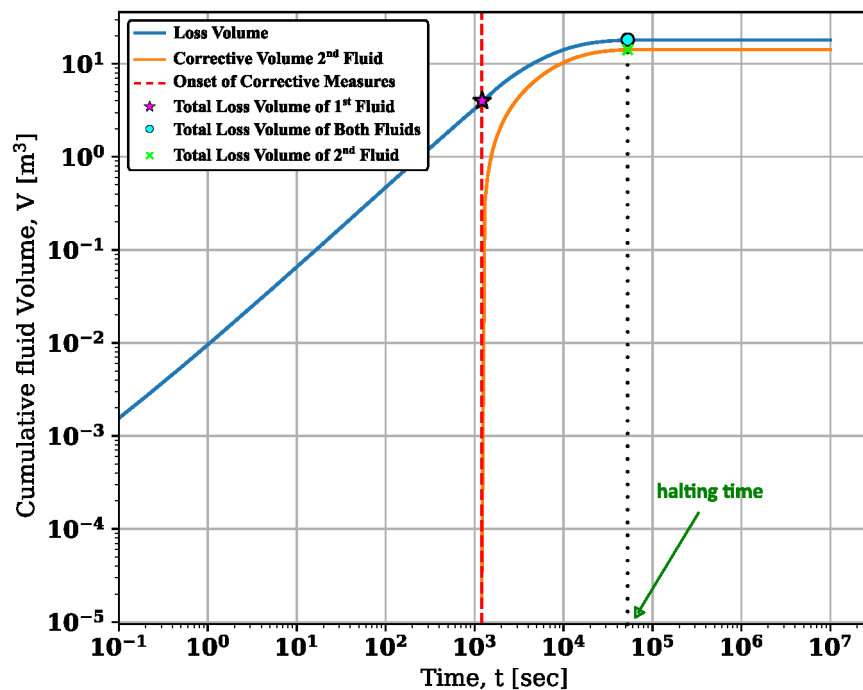


Figure 3—2D plot for a selected scenario of LC and its corrective measures. The y-axis represents the elapsed time since LC occurs, and y-axis presents the cumulative fluid loss volume.

Another applicability of the solution is the exhibition of a case of no corrective measures that were considered, and the LC event is allowed to continue escaping into the fracture. Sequentially, the loss severity will be higher, and no mitigation will be observed. Fig. 4 provides the scenarios of no introduction of corrective fluid and LC exhibits higher cumulative loss volume and higher halting time. Given the identical 1st fluid properties and other parameters presented priorly in Fig. 3, the total cumulative loss volume is calculated to be 63.3 $[m^3]$ corresponding to stopping time estimated to be 80 [hrs]. Average fracture aperture of 0.72 [mm], wellbore radius of 3.6 [in] and pressure drop of 100 [psi] are used during the demonstration process. Table 1 shows a summary of the study of the two schemes for the same case. The monitoring parameters are the criteria for the key performance indicators of the model, and the last one is the used fluid properties. The physical parameters are, total cumulative loss volume at stopping conditions, ceased time whenever LC completely halts, final cumulative lost volume at stopping conditions for 1st and 2nd fluids. The comparison is made between LC in addition to corrective measures and without the measures.

From the table below, there is a significant improvement in the total lost volume reported with corrective measures. By merely modifying two fluid parameters slightly, a mitigation is obviously noted for corrective measure scheme. Also, an earlier cease of LC is noticed with great improvement for the same scheme. This table represents a show case for only one scenario as varying input parameters can produce a broad range of results.

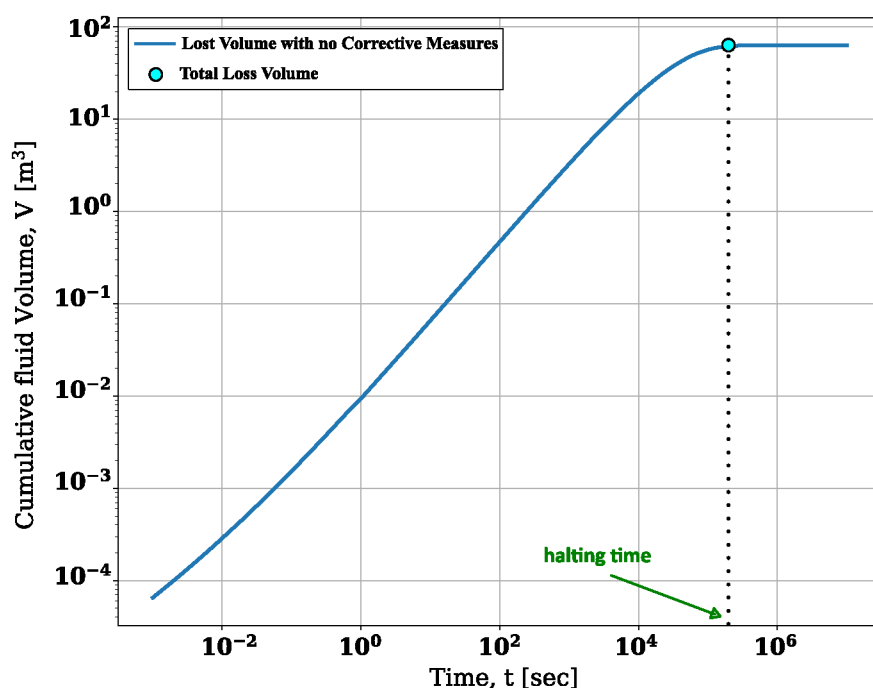


Figure 4—2D plot for LC with no corrective measures. The y-axis shows cumulative fluid loss volume and x-axis is the duration of the LC incident.

Table 1—Two cases demonstrate the applicability of corrective measures and their impact on mitigation of LC severity.

Monitoring parameter	LC with corrective measure	LC with no corrective measure
Total cumulative loss volume	18.1 [m³]	63.3 [m³]
Halting time	15.5 [hrs]	80 [hrs]
Total 1 st fluid lost volume	3.8 [m³]	63.3 [m³]
Total 2 nd fluid lost volume	14.3 [m³]	--
Fluid properties	1 st fluid: $n=0.94, m=0.01 [Pa.s^n], \tau_{0-1} [Pa]$ 2 nd fluid: $n=0.89, m=0.01 [Pa.s^n], \tau_{0-2} [Pa]$	1 st fluid: $n=0.94, m=0.01 [Pa.s^n], \tau_{0-1} [Pa]$

Field Case Application

A field case will be shown to demonstrate the feasibility of the established model. A drilling fluid loss was encountered in Gulf of Mexico (GOM) in a zone (Z1) from 6:22 to 7:20 a.m. in a region prone to LC at depth around 28,070 [ft], operating under British Petroleum company. The raw dataset in volumetric flow rate in gallon per minute is represented in Fig. 5. The fluid properties reported for this incident are: $\tau_0 = 8.4 [lb/100ft^2]$, $m=0.08 [lb/lb/100ft^2]$ and $n=0.94$. The process undergoes firstly a transformation of the clock-time to duration time in [sec]. Then, cumulative loss volume is calculated based on Eq. (22), where it is plotted in y-axis over duration time as presented in Fig. 6. Afterwards, the matching solution is provided under the assumption of one fluid infiltrating into the natural fracture. This fluid continues

to propagate gradually with no action taken until a complete stop is reached, i.e. no corrective fluid is introduced, as demonstrated in Fig. 6-top. Considering corrective measures scenario, a 2nd fluid is addressed into the LC phenomenon at 0.42 [h] that mitigates the loss significantly in terms of loss volume and halting time, portrayed in Fig. 6-down.

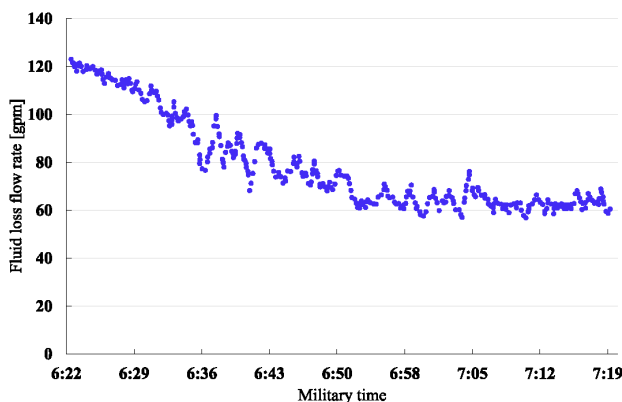


Figure 5—LC incident reported by BP as lost fluid volumetric flow rate in gallon per minutes over clock hours (24-format) reported.

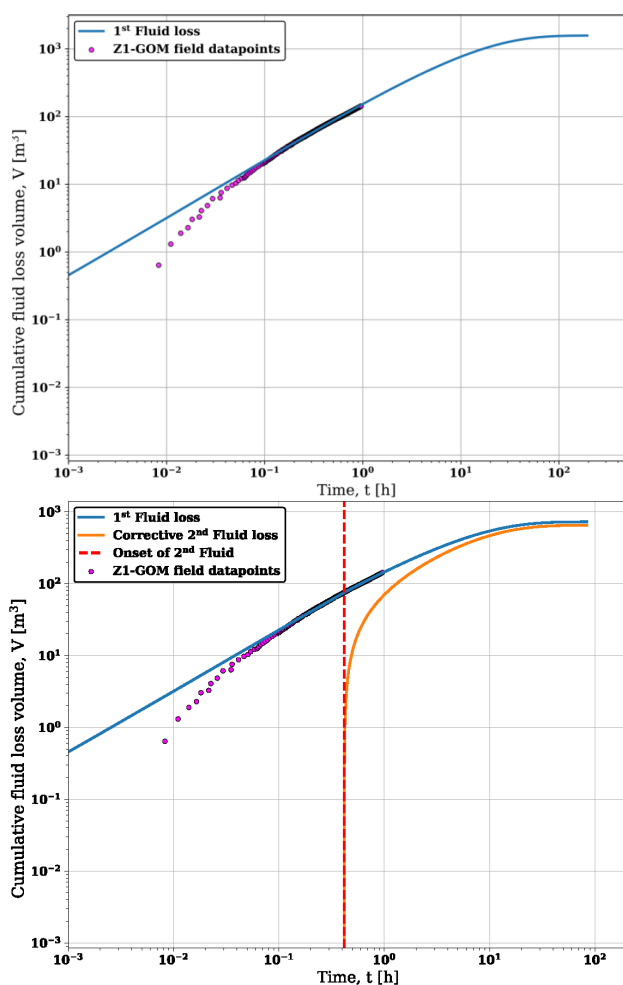


Figure 6—Modeling application of a field case of GOM-Z1 showing the effect of a corrective fluid scenario (bottom) comparing with non-corrective scenario (top).

Discussion

This study presents a modeling for LC incident into a natural fracture assuming a horizontal one. Yet, the complexity of the natural fracture system is simplified and represented by a single horizontal one for practical and simplification purposes. Approaching a complex problem, such as LC, cannot afford a long-time consuming approach, for instance full numerical simulation, since the LC may happen unexpectedly and the solution is demanded to be ready within an extremely short amount of time. Thus, simplifications of LC in a natural fracture were assumed to be horizontal and single body one as a representation of a complex fracture network. Another important parameter for analysis is fluid front radius. This represents an equivalent fluid front as the radius follows a perfect circular shape along azimuth. Indeed, there are variations of fluid front radii along azimuth pole and tracking the exact shape of fluid invasion over radial axis is therefore of no interest in this study as the goal is to monitor the fluid loss volume dynamically. This work applies merely if the cause of LC originates from the presence of a conductive natural fracture, because there are other factors causing LC, such as vugs, caverns, high permeable porous zone and induce fracture.

Roughness is a critical factor in determining the shape of fracture aperture surface map. The scope of the work is not focusing on roughness determination for the mechanical aperture of the fracture, but rather a representative value, i.e. average hydraulic fracture aperture. In this work, the assumption of an average fracture aperture is utilized to perform the calculation of dynamic cumulative lost volume. Indeed, the shape of the fracture with its characteristic factors of roughness may add a higher volume estimate. However, this increase in volume will be compensated and reflected by the determination of fracture aperture as it is an inverse problem.

This study aids also a guideline for selection process of Lost-circulation materials (LCM) because it can estimate the average hydraulic fracture. LCM is used based on selection criteria of fracture aperture range, as supported previously with uncertainty quantification analysis [13]. Based on another previous published work by generating type-curves, average hydraulic fracture aperture can be swiftly estimated onsite or at real-time operation center [11].

Conclusion

LC is an issue facing the industry with often no expectation of its occurrence. Once LC is encountered, corrective measures are taken place to mitigate the severity of the loss. In this paper, modeling the corrective measures as introducing a plugging fluid, referred in this paper as 2nd fluid, is presented along with its mathematical derivation of two-phase non-Newtonian fluid flow following Herschel-Bulkley fluid, where corrective fluid displacing the lost fluid. The corrective fluid serves as a thicker fluid causing a slow down of fluid loss and ultimately reaching to stop, due to shear thickening behavior. LCM can be added to the corrective fluid to alter its fluid properties in favor of contributing to the mitigation action. Wide range of scenarios can be generated by varying fluid properties of both fluids, average fracture aperture, pressure drop and wellbore radius size and "what-if" analysis can be easily performed. Only one case was focused on showing the feasibility of the model by monitoring the lost volume over time. Moreover, a scheme was generated for LC with no corrective measures. This was done by allowing LC to continue and monitoring the cumulative loss volume dynamically. These two schemes were compared and analyzed. The benefits of this work are to;

- Comprehend the behavior of non-Newtonian fluid
- Determine the controlling factors of LC, such as fluid properties, fracture size, wellbore radius and pressure difference
- Indicate the factor controlling the stopping condition of the LC
- Assist the design of corrective fluid to achieve the desirable mitigation level of severity

- Reveal the effect on swift decision-making on LC incident when introducing a corrective fluid, referring to the onset time of a 2nd fluid
- Support the capabilities of real-time operation center to tackle LC problem

References

1. A. K. Al-Delfi and F. H. M. Al Mahdawi, "A comprehensive review of lost circulation: Principles and treatments," 2023, p. 020026. doi: [10.1063/5.0167937](https://doi.org/10.1063/5.0167937).
2. C. Carpenter, "Lost Circulation Simulated Accurately in Primary Cementing-Job Design," *Journal of Petroleum Technology*, vol. 77, no. 05, pp. 1–4, May 2025, doi: [10.2118/0525-0025-JPT](https://doi.org/10.2118/0525-0025-JPT).
3. Y. Sun et al, "Development and Performance Evaluation of Ultralow-Temperature-Resistant Plugging Agent for Antarctic Drilling," *SPE Journal*, pp. 1–10, May 2025, doi: [10.2118/226208-PA](https://doi.org/10.2118/226208-PA).
4. F. AlSayed Omar et al, "Advanced Lost Circulation Solution: A Highly Thixotropic Cement System Pumped through BHA for Efficient Lost Circulation Management," in *APOGCE 2024*, SPE, Oct. 2024. doi: [10.2118/221257-MS](https://doi.org/10.2118/221257-MS).
5. S. Ashoori, M. Bahari Moghadam, R. Nazemi, V. Nooripoor, and M. Ahmadabadi, "Dynamically Evaluating the Performance of Naturally Occurring Additives to Control Lost Circulation: On the Effect of Lost Circulation Material Type, Particle-Size Distribution, and Fracture Width," *SPE Journal*, vol. 27, no. 05, pp. 2590–2612, Oct. 2022, doi: [10.2118/209620-PA](https://doi.org/10.2118/209620-PA).
6. S. Savari, J. Butcher, and M. Al-Hulail, "Managing Lost Circulation in Highly Fractured, Vugular Formations: Engineered Usage of High Fluid Loss Squeeze and Reticulated Foam Lost Circulation Materials," in *IADC/SPE International Drilling Conference and Exhibition*, SPE, Feb. 2020. doi: [10.2118/199635-MS](https://doi.org/10.2118/199635-MS).
7. L. Pu, P. Xu, M. Xu, J. Song, and M. He, "Lost circulation materials for deep and ultra-deep wells: A review," *J Pet Sci Eng*, vol. 214, p. 110404, Jul. 2022, doi: [10.1016/j.petrol.2022.110404](https://doi.org/10.1016/j.petrol.2022.110404).
8. P. Sarmah, "High-Density High-Strength Phase Transforming Lost Circulation Material to Cure Severe to Total Losses in Reservoir and Non Reservoir Sections," in *Middle East Oil, Gas and Geosciences Show*, SPE, Mar. 2023. doi: [10.2118/213746-MS](https://doi.org/10.2118/213746-MS).
9. F. Xiao et al, "Reducing Mud Loss in Drilling Operations with High-Performance Plugging Agent," in *Mediterranean Offshore Conference*, SPE, Oct. 2024. doi: [10.2118/223230-MS](https://doi.org/10.2118/223230-MS).
10. K. Nguyen, A. Mehrabian, A. Santra, D. Phan, and A. Bathija, "Lost Circulation Material Size Selection for Naturally Fractured Rocks," in *Day 3 Wed*, February 14, 2024, IPTC, Feb. 2024. doi: [10.2523/IPTC-24460-MS](https://doi.org/10.2523/IPTC-24460-MS).
11. R. Albattat and H. Hoteit, "A semi-analytical approach to model drilling fluid leakage into fractured formation," *Rheol Acta*, vol. 60, no. 6-7, pp. 353–370, Jul. 2021, doi: [10.1007/s00397-021-01275-3](https://doi.org/10.1007/s00397-021-01275-3).
12. K. Nguyen, A. Mehrabian, A. Santra, D. Phan, and A. Bathija, "Lost Circulation Material Design for Highly Inclined Or Horizontal Well Sections," in *Middle East Oil, Gas and Geosciences Show*, SPE, Mar. 2023. doi: [10.2118/213698-MS](https://doi.org/10.2118/213698-MS).
13. R. Albattat, M. AlSinan, H. Kwak, and H. Hoteit, "Modeling lost-circulation in natural fractures using semi-analytical solutions and type-curves," *J Pet Sci Eng*, vol. 216, p. 110770, Sep. 2022, doi: [10.1016/j.petrol.2022.110770](https://doi.org/10.1016/j.petrol.2022.110770).
14. R. Albattat, X. He, M. AlSinan, H. Kwak, and H. Hoteit, "Modeling Lost-Circulation in Fractured Media Using Physics-Based Machine Learning," in *83rd EAGE Annual Conference*

- & Exhibition, European Association of Geoscientists & Engineers, 2022, pp. 1–5. doi: [10.3997/2214-4609.202210204](https://doi.org/10.3997/2214-4609.202210204).
15. K. Tsusaka et al, "An Integrated Geomechanical Approach in Mud Loss Pressure Prediction in a Giant Oil Field in Offshore Abu Dhabi," in ADIPEC, SPE, Oct. 2023. doi: [10.2118/216715-MS](https://doi.org/10.2118/216715-MS).
 16. K. Nguyen, A. Mehrabian, A. Santra, D. Phan, and A. Bathija, "Lost Circulation Material Design for Engineered Fracture Gradient of Drilling," in SPE Annual Technical Conference and Exhibition, SPE, Sep. 2022. doi: [10.2118/210027-MS](https://doi.org/10.2118/210027-MS).
 17. T. Saihood and R. Samuel, "Mud Loss Prediction in Realtime Through Hydromechanical Efficiency," in ADIPEC, SPE, Oct. 2022. doi: [10.2118/211777-MS](https://doi.org/10.2118/211777-MS).
 18. M. I. Okai, O. Ogo, P. Nzerem, and K. S. Ibrahim, "Application of Boosting Machine Learning for Mud Loss Prediction During Drilling Operations," in SPE Nigeria Annual International Conference and Exhibition, SPE, Aug. 2024. doi: [10.2118/221583-MS](https://doi.org/10.2118/221583-MS).
 19. H. Pang, H. Wang, Y. Jin, Y. Lu, and Y. Fan, "Prediction of Mud Loss Type Based on Seismic Data Using Machine Learning," in 56th U.S. Rock Mechanics/Geomechanics Symposium, ARMA, Jun. 2022. doi: [10.56952/ARMA-2022-0485](https://doi.org/10.56952/ARMA-2022-0485).
 20. Y. Feng, H. Yang, X. Li, S. Zhang, H. Hu, and J. Wang, "Interpretable Lost Circulation Analysis: Labeled, Identified, and Analyzed Lost Circulation in Drilling Operations," *SPE Journal*, vol. **29**, no. 04, pp. 1692–1709, Apr. 2024, doi: [10.2118/218380-PA](https://doi.org/10.2118/218380-PA).
 21. H. H. Alkinani et al, "Economic Analysis of Lost Circulation Events to Optimize the Drilling Process in Basra Oil Fields, Iraq," in *Day 2 Wed*, March 27, 2019, IPTC, Mar. 2019. doi: [10.2523/IPTC-19290-MS](https://doi.org/10.2523/IPTC-19290-MS).
 22. C. Winn et al, "Context and mitigation of lost circulation during geothermal drilling in diverse geologic settings," *Geothermics*, vol. **108**, p. 102630, Feb. 2023, doi: [10.1016/j.geothermics.2022.102630](https://doi.org/10.1016/j.geothermics.2022.102630).
 23. R. Albattat and H. Hoteit, "Novel Analytical Solution and Type-Curves for Lost-Circulation Diagnostics of Drilling Mud in Fractured Formation," in SPE Middle East Oil & Gas Show and Conference, SPE, Dec. 2021. doi: [10.2118/204619-MS](https://doi.org/10.2118/204619-MS).
 24. R. Albattat and H. Hoteit, "Modeling yield-power-law drilling fluid loss in fractured formation," *J Pet Sci Eng*, vol. **182**, p. 106273, Nov. 2019, doi: [10.1016/j.petrol.2019.106273](https://doi.org/10.1016/j.petrol.2019.106273).